

Irrationality of $F(31/4)$ and the Exact Normalized Hankel Order

A rational base outside the Bundschuh–Väänänen region, conditional on two lemmas of Zudilin, with the sharp q -order and the arithmetic obstructions at $3/2$ for Erdős Problem #1049

WILL COOK*

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ABSTRACT

Let

$$F(t) = \sum_{n \geq 1} \frac{1}{t^n - 1}.$$

Erdős Problem #1049 asks whether $F(t)$ is irrational for every rational $t > 1$. The problem remains open, and this paper proves no irrationality result at $t = 3/2$.

At one rational base it proves irrationality outright. Zudilin’s q -harmonic construction in the direction $(14, 12, 14; 27)$ survives substitution of a rational base directly into its polynomial identity, and the resulting exponent budget gives the threshold $\log b / \log a < C_0/C_1 = 0.4056830213840605\dots$ for coprime $a > b \geq 1$. Since $\log 4 / \log 31 < 81/200$ lies below it, $F((31/4)^r)$ is irrational for every $r \geq 1$. Zudilin announced in 2016 that his results extend to a rational base under an assumption of exactly this shape, for some computable constant he did not name; what is supplied here is that constant, and with it the strip $(0.398678, 0.405683]$ above the Bundschuh–Väänänen region, which contains $31/4$ and no integer base. The proof consumes Lemmas 1, 2 and 7 of Zudilin 2004 and nothing else external. It is not kernel-checked; Lean checks the finite part, namely the trigamma constants, the bounds $81/200 < \theta^* < 1/2$, and the membership and exclusion of the bases involved. A companion Archimedean cap shows that every Padé-type family of this shape is confined to $b^2 < a$, so the same mechanism reaches no base with $b^2 \geq a$, $3/2$ included.

Our strongest unconditional theorem sharpens the normalized Hankel estimate in Zudilin’s Padé construction. At $x = z = 1$, if V_N^* is his rank- N normalized determinant, then for every $N \geq 1$

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6}, \quad [q^{\text{ord}_q V_N^*}] V_N^* = \frac{(N!)^2 (N+1)!}{2^N}.$$

The published argument proves the displayed order only as a lower bound. An associated-graded recurrence identifies every transformed-row leading coefficient, and a Vandermonde determinant proves equality. Thus additional normalized-Hankel q -order cannot supply the missing analytic decay in this specific construction. This is a route closure,

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not an irrationality result: denominator-factor extraction and alternative integral models remain open.

Separately, at a rational target a/q , prime support in the denominator coordinate of an integral approximation row forces the sharp gap $|Ba/q - A| \geq 1/q$. A first-power determinant condition forces that gap in at least one of two rows, and two noncollinear integral rows cannot both have remainders tending to zero. These are exact arithmetic endgames, but their prime-support and analytic-decay hypotheses are not constructed for the actual q -Apéry family.

At $3/2$, a second exact input is

$$2^{64} < 3^{41} < 2^{65}.$$

It yields a uniform gap greater than $3/13$ between $\log 2 / \log 3$ and every threshold in the rectangular Hermite–Padé exponent model considered here. It also shows that the scalar factor, and the scalar factor together with the universally forced first-order border, fall short of the required $39/41$ charge. These inequalities exclude the stated height mechanisms only; they do not exclude other Padé or determinant constructions.

The same bracket sharpens a finite four-jet count. At bottom depth $R = 41T$, any $M \geq 130T + 2S$ integral polynomial pairs have two binary selectors with the same two endpoint jets in both coordinates. For $T = 1$ this counting threshold is exact. A bounded-fibre lemma shows how a uniform bound on analytic-remainder multiplicities would force one collision outside the remainder nullspace. No such bound is proved for an actual q -Apéry or Zudilin family. Under the additional hypothesis that adjacent minors vanish and the second coordinates are units, a Bézout shear reduces the sufficient selector width from $2S + 4R$ to $S + 2R$.

Two further obstructions explain why direct integer-base arguments do not transfer automatically to $3/2$. Integer scalar content scales local divisibility and Archimedean determinant height by the same factor, while unit endpoint coefficients keep both 2 and 3 out of a common divisor after specialisation. A six-condition coordinatewise clearing scheme forces $s^{N+K+1} < r(N+K)$; at $r/s = 3/2$ this is impossible for every $N, K \geq 1$. The exact cleared-tail recurrence contains the forcing term $Bc(N+1)s^{N+1}$, which is at least 2^{N+1} when $s \geq 2$ and the integer data are positive.

For comparison, Bundschuh and Väänänen’s published criterion proves irrationality for a restricted rational-base family containing $7/2$; only its elementary height check is formalised here. The Padé exponent bounds and the $81/200$ logarithmic-region calculations below likewise supply no analytic remainder estimate at $3/2$. The unresolved task is to construct a primitive, noncollapsed approximation family whose endpoint divisibility survives normalisation, whose selected polynomial pair and remainder are nonzero, and whose decay beats its height.

Sharp formula and exact remaining boundary

All-rank formula. Zudilin’s normalized Hankel lower bound is an equality at $x = z = 1$, with first coefficient $(N!)^2(N+1)!/2^N$; hidden normalized-Hankel order is therefore unavailable as extra analytic decay. **Rational gap.** Prime-supported integral rows have a sharp $1/q$ gap, and noncollinear pairs cannot both decay to zero; constructing the required rows remains open. **Exact arithmetic.** The bracket $2^{64} < 3^{41} < 2^{65}$ excludes the stated scalar and first-order Hermite–Padé height mechanisms at $3/2$. It also yields the sharp finite four-jet count at bottom depth 41, with a Bézout–Plücker reduction under explicit minor and unit hypotheses. **Transfer obstruction.** Direct integer-base clearing fails by an exact growth inequality and forcing recurrence. **Open boundary.** No primitive noncollapsed approximation family with a sufficiently small nonzero remainder is constructed, so no irrationality at $3/2$ is proved.

1 Introduction

Let $t > 1$ be a rational number and let $\tau(n)$ count the divisors of n . Erdős Problem #1049 asks whether

$$F(t) = \sum_{n \geq 1} \frac{1}{t^n - 1} = \sum_{n \geq 1} \frac{\tau(n)}{t^n}$$

is irrational [2, p. 102].

The two forms agree by expanding $(t^n - 1)^{-1} = \sum_{k \geq 1} t^{-nk}$ and collecting the terms with the same exponent, the coefficient of t^{-m} being the number of divisors of m . The question is a conjecture of Chowla; Erdős proved it for every integer $t \geq 2$ [1]. Bloom’s current catalogue record reproduces the displayed rational- t question, labels it *open*, attributes it to Chowla, and points to Erdős’s 1988 statement on p. 102 and the 1948 integer-base theorem [19]. The same record warns that its status is the website owner’s current assessment and asks readers to cite the original Erdős sources. Accordingly, the catalogue is used here for numbering and current reported status, while the two original publications carry the mathematical claims. The universal conjecture over all rational $t > 1$ remains open; individual non-integral rational bases are known, including $7/2$ below.

Write $t = r/s$ in lowest terms with $r > s \geq 1$, so that $s = 1$ is exactly the integer case Erdős settled. The resistant explicit base of least naive height $H(r/s) = \max(r, s)$ is $t = 3/2$. A published height criterion of Bundschuh and Väänänen [6, Thm. 2, p. 177; hypotheses pp. 175–176] settles a family of rational bases restricted by a height condition; that family contains $7/2$ and does not contain $3/2$. Between the two lies the question this note is about: what exactly stops the integer-base argument from running at $3/2$?

Relation to prior work. The *Formal Conjectures* file for Problem #1049 contains the conjecture and integer-base theorem as sorry placeholders, but it also proves the Lambert identity between the two displayed series for rational t [20]. In the present $t > 1$ regime its `lambert_convergent` branch is an ordinary convergent-series proof; the same file’s `|t| ≤ 1` branch instead uses Lean’s convention that the `tsum` of a nonsummable series is zero. Thus it supplies genuine checked prior art for the identity, but no irrationality

theorem. The present note formalises propositions about the clearing argument and likewise does not answer the conjecture.

Erdős's positive-integer theorem of 1948 [1] sits inside a larger integer-base literature. At the level of functions, Rivin proves that if a sequence γ and its divisor-sum sequence are both eventually linearly recurrent, then γ is finitely supported [17, Theorem 1.1, p. 2; proof pp. 6–7]. His periodic-coefficient corollary therefore shows that

$$\sum_{n \geq 1} \frac{z^n}{1 - z^n}$$

is not a rational function [17, Corollary 6.4, p. 9]. This is an exact structural statement about the function underlying Problem #1049, but it gives no irrationality statement for a special value at $z = 1/t$: a nonrational function may take rational values at particular rational points.

In 1991 Borwein proved the irrationality of shifted series $\sum_{n \geq 1} (t^n + w)^{-1}$ at integer bases $t \geq 2$, for every nonzero rational w with $w \neq -t^m$ for all $m \geq 1$, by Padé approximation rather than by digit clearing; his estimates also show that these values are not Liouville numbers [3, Thm. 4, pp. 257–258]. In 2001 Van Assche recovered the integer-base irrationality and the bound $\mu(F(p)) \leq 2\pi^2/(\pi^2 - 2) = 2.50828 \dots$ using little q -Legendre Padé approximants [14, Thm. 1, p. 10; proof pp. 10–11]. His more general Theorem 3 proves irrationality of $\sum_{k \geq 1} (cp^k - 1)^{-1}$ for an integer $p > 1$ and fixed rational c away from the poles [14, Thm. 3, p. 14]; it does not cover a rational noninteger base t in $F(t)$, because the multiplier needed to write t^k over an integer base varies with k .

The same Lambert value was already the target of Amdeberhan and Zeilberger's q -WZ construction [5]. Here $p > 1$ is the integer-base parameter and the little- q kernel uses $q = p^{-1}$. The two constructions share that bivariate little- q -Legendre Padé kernel, but Van Assche's diagonal does not satisfy the Amdeberhan–Zeilberger scalar recurrence. Their Theorems 1 and 2 do prove, respectively, the irrationality of the non-alternating $h_p(1)$ and alternating q -logarithm values for their stated integer parameters, with reported irrationality measure 4.80; those are external integer-parameter results and do not transfer to the rational noninteger base $3/2$ studied here.

Amdeberhan–Zeilberger use the moving diagonal $P_n(p^{n+1} \mid p^{-1})$, whereas Van Assche uses $P_n(p^n \mid p^{-1})$. Van Assche also records, citing Borwein's 1992 Lemma 2, the neighbouring evaluation $P_{n-1}(cp^{n+1} \mid p^{-1})$ [4, 14]. The shared kernel therefore does not license transfer of recurrence, endpoint, lattice, or valuation claims between the diagonals. Indeed, if $A_n(p) = P_n(p^n \mid p^{-1})$, exact substitution at $n = 0$ into the Amdeberhan–Zeilberger operator leaves

$$-p(p-1)^2(p+1)(p^5 + 2p^4 + 2p^3 + 2p^2 + 2),$$

which is nonzero for every real $p > 1$. One nonzero residual is decisive for non-transfer of that recurrence.

Lean checks the [exact residual factorisation](#) and its [nonvanishing for \$p > 1\$](#) . These are finite statements at $n = 0$; they do not supply a recurrence for either diagonal. No recurrence, endpoint, lattice, or valuation statement for one diagonal is used for the other; the displayed exact residual is the sole claim made here about their incompatibility.

There is a second distinction at rational base $3/2$. Over the checked range, the raw rational approximation errors decrease while the selected cleared integer forms grow from $n \geq 2$. This says that the chosen integerisation does not produce small linear forms; it does *not* say that the raw approximants fail to converge.

In 2013 Vandehey proved that $\sum_{n \geq 1} d(n)a_n/b^n$ is irrational whenever $b > 1$ is an integer and (a_n) ranges in a finite integer alphabet excluding zero; taking $a_n = (-1)^n$ completes the elementary digit method for integer bases $b \leq -2$ [15, Thm. 1.2, p. 2]. His companion theorem permits a finite alphabet of nonnegative integers containing zero, provided the coefficient sequence is not eventually zero [15, Thm. 1.1, p. 2]. All three retain an integer base. Luca and Tachiya's periodic-coefficient theorem likewise proves irrationality at every integer base of absolute value greater than one; their full-support example strengthens this to joint linear independence for finite families of iterated divisor functions [16, Theorem A, p. 139; Example 1, p. 140]. This is strong integer-base evidence, but it does not cover a rational noninteger base.

In 2004 Zudilin obtained the uniform bound $\mu(F(p)) \leq 2.46497868 \dots$ for every integer $p \notin \{0, \pm 1\}$, by way of Heine's basic transform and a permutation group [8, Thm. 1, p. 154; Secs. 4–5, pp. 159–162]. The ordinary-hypergeometric antecedent is Rhin and Viola's S_5 action and twelve-coset denominator reduction for rational forms in $\zeta(2)$ [13, Sec. 3, pp. 38–42; Sec. 4, pp. 46–51].

Those sources motivate the permutation, denominator and polynomial- specialisation architecture of Section 7; the endpoint lemmas there are abstract and are not yet applied to either source's actual coefficient family.

The neighbouring problem of Lambert subseries $\sum_{n \in A} (t^n - 1)^{-1}$ over a restricted index set A is treated by Kovač and Tao [18].

The rational non-integer progress relevant here is instead the height criterion of Bundschuh and Väänänen [6], which remains the strongest explicit numerical threshold among the sources compared here for this value. Zudilin's later Padé-and-Hankel treatment also admits non-integral rational bases: its generalized q -logarithm remarks that its results also hold for non-integer $p = r/s$ with $|p| > 1$, under an assumption $\log |r| > c \log |s|$ for some computable $c > 0$ [9]. No value of c is computed there. Section 2 computes a threshold of exactly that shape, $\log s / \log r < C_0/C_1$, from the earlier direction [8] rather than from the generalized q -logarithm. The remark is therefore both the closest antecedent and the sharpest reason to withhold any priority claim: the rational-base extension was contemplated in print, only its constant was left uncomputed.

Section 3 returns to the normalized determinant in Zudilin's Section 4 and sharpens its q -order estimate at $x = z = 1$. The closest recent determinant comparison we found is Chern–Jiu–Li–Wang's theory of q -binomial transforms [11]. Their theorems determine the highest degree and leading coefficient in an independent transform parameter for two finite q -binomial transforms of an arbitrary sequence. The result below instead determines the lowest q -exponent and first coefficient of Zudilin's particular infinite q -hypergeometric moment sequence after fixing $x = z = 1$. Their paper neither treats this sequence nor the generalized q -logarithm, so its leading-coefficient theorems do not imply Theorem 3.1. This is an exact comparison of theorem statements, not a

priority claim. No claim of priority is made for the analysis below. It records the formal status and limits of the argument; it does not add a theorem about $F(t)$.

Erdős's integer-base argument, in outline. For an integer $b \geq 2$, Erdős rewrites $F(b) = \sum_{n \geq 1} \tau(n)b^{-n}$. A Chinese-remainder construction forces arbitrarily long blocks in which the divisor coefficients have the powers of b needed to make the corresponding base- b digits zero. Explicit bounds control the middle and far tails, while positivity proves that the expansion does not terminate. The resulting base- b expansion has arbitrarily long zero blocks without being eventually zero and is therefore irrational [1, pp. 63–66].

The direct cut-and-clear attempt studied here. A more naive attempt is to suppose $F(b) = p/q$, cut at N , and multiply the remaining identity by qb^N . This does not by itself trap a positive integer below 1: the first uncleared tail contribution is $q\tau(N+1)/b$. The sections below isolate additional corridor hypotheses under which a bounded-window version of this attempt would work, and then show why those hypotheses fail at $3/2$.

Write $\beta = r/s$ for the base and $c(n)$ for the coefficient of β^{-n} , so that $c = \tau$ in the case at hand. The term $c(n)\beta^{-n}$ is $c(n)s^n/r^n$. Clearing the power of r leaves the numerator factor s^n in place. That factor is invisible when $s = 1$ and grows geometrically when $s \geq 2$. At $3/2$ it is 2^n .

Terminology. The linear-form method uses polynomial pairs before specialisation and integer pairs afterwards. We reserve *row* for an integer pair (U, V) . Its *integer scalar content* is $\gcd(|U|, |V|)$; a row is *primitive* when this number is 1, and dividing by it is *primitive normalisation*. A polynomial pair may instead have a polynomial common factor in $\mathbb{Z}[X]$; that is a different operation and is not called row content here. The *exterior determinant* of two integer rows is $U_n V_m - U_m V_n$, the determinant of the 2×2 matrix they form. Two quantities attached to that determinant are compared throughout: an integer dividing it, which is a local gain, and its absolute value, which is an Archimedean cost; we call that comparison the *local-to-Archimedean balance*.

The *endpoints* of a coefficient polynomial, taken relative to the declared width W of Section 7, are its constant coefficient and its coefficient at W ; we call these the *constant endpoint* and the *top endpoint*, so the top endpoint is the coefficient at W and not the leading coefficient unless the two agree. A *unit endpoint* is one equal to ± 1 , the units of $\mathbb{Z}$; Theorem 7.1 is the reason only these two coefficients decide divisibility by 3 and by 2 after specialisation at $(3, 2)$. A *jet* is a residue of a specialised coefficient modulo a prime power: the bottom jet is its residue modulo 3^R and the top jet its residue modulo 2^S , and R and S are the bottom and top *depths*. A jet vanishes exactly when the prime power in question divides the specialised coefficient.

The shortfall at $3/2$. The elementary method and the linear-form method are both examined below. For the height argument, the exact bracket $2^{64} < 3^{41} < 2^{65}$ gives a gap greater than $3/13$ across the whole admissible rectangular exponent cone and fixes the $39/41$ denominator-charge comparison. Neither the source scalar factor nor that factor together with the universally forced first-order residual border reaches the required charge. These comparisons leave higher residual valuations, determinant cancellation,

and other integral models untouched. For the additive argument, the same bracket replaces the generic $4R + 2S$ threshold by $130T + 2S$ at depth $R = 41T$ and gives an exact first-row failure at $T = 1$.

For the coordinatewise clearing scheme the leftover at each step is the forcing term of an exact recurrence, of size at least 2^{N+1} whenever $s \geq 2$ and the scaling constant B and the coefficient $c(N+1)$ are at least 1 (Theorem 9.2), and the scheme itself is excluded at $3/2$ for every shift $N \geq 1$ and every cleared window of width $K \geq 1$ (Theorem 8.5).

For the linear-form constructions we examine two possible sources of 2-adic and 3-adic gain. Integer scalar content is exactly neutral, since it scales the exterior determinant and its absolute height by the same factor (Theorem 6.1), while unit endpoints keep both 2 and 3 out of any common divisor of the two specialised evaluations (Theorem 7.1 and Proposition 7.3). Corollary 7.5 states the two scoped exclusions together; neither is claimed to be necessary for every linear-form proof. Separately, the scalar parameter margin is negative under the assumed source inequality (Theorem 7.12).

One candidate pursued here is additive: an integer relation among rows that cancels the endpoint jets. Theorem 7.7 shows that a nonzero relation with coefficients in $\{-1, 0, 1\}$ cancelling all four jets exists whenever the bottom depth is positive and the number of coefficient pairs is at least $4R + 2S$. It does not show that the resulting combination has a nonzero polynomial pair or a nonzero remainder. Problem 12.1 gives a precise sufficient specification for that particular candidate architecture, not a necessary condition for solving Problem #1049.

Sections 10 and 11 record two external methods and what each leaves unproved at $3/2$.

Sharpness. Two questions of scope are worth isolating. The corridor bound of Theorem 8.2 is exponential on the left and linear on the right, so for a fixed numerator and any $s \geq 2$ a corridor can survive only for bounded $N + K$; the base $3/2$ is the case in which the crossing has already happened at the smallest admissible window, which is why the exclusion there holds for all $N \geq 1$ and $K \geq 1$ with no further restriction. The height criterion of [6] is restricted by a height condition satisfied at $7/2$ and not at $3/2$, so the two bases are separated by that criterion rather than by a universal obstruction. The exponent 65 in $3^{41} < 2^{65}$ cannot be replaced by 64. The separate direct comparison $2^{129} < 3^{82}$ then proves that the integer four-jet threshold is exactly $130 + 2S$ when $R = 41$: $129 + 2S$ does not suffice.

For $R = 41T$ with $T > 1$, the displayed $130T + 2S$ bound is uniform and sufficient, but no claim of exact optimality at every T is made.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

Results and boundary. The linked Lean declarations establish exact inequalities, congruences, finite collision counts, recurrence identities, and exclusions for the named models. The normalized-Hankel theorem is an ordinary all-rank proof with a Lean-checked source-tail, recurrence, and determinant chain plus finite replay, not yet an end-to-end Lean theorem. Bundschuh and Väänänen’s theorem at $7/2$, Rivin’s functional nonrationality theorem, and the two-variable Mahler theorem used below remain external results. None of the formal statements proves an irrationality result at a rational noninteger base. At $3/2$, the missing step is still an actual primitive approximation family with nonzero remainder and an asymptotic height margin.

Structure. Section 3 proves the all-rank normalized-Hankel order and leading coefficient and states exactly which analytic route it closes. Section 4 derives the height and charge exclusions from the sharp $41/65$ power certificate. Section 5 proves the sharp one- and two-row rational gaps, the no-double-decay theorem, the real unit-determinant height-decay tradeoff, and the reduced selector collision. Section 6 proves that integer scalar content is neutral for the local-to-Archimedean balance.

Section 7 proves the endpoint congruences at $(3, 2)$, deduces from them and from Section 6 that neither integer scalar content nor a common divisor of the two specialised evaluations supplies the targeted endpoint gain, gives the four-jet collision count and its conditional Bézout–Plücker compression, and records one further exclusion on Zudilin’s scalar parameters. Sections 8 and 9 return to the elementary clearing scheme and record what the residue s^n costs there, first as an exclusion and then as an exact recurrence with a lower bound on the surviving term. Sections 10 and 11 record what is and is not formalised of two external routes, together with the separate $81/200$ logarithmic comparison. Section 12 separates kernel escape from asymptotic adequacy and states the remaining obligations. Linked phrases open the corresponding Lean declaration at the pinned source revision `ca0e13f8acf5`.

Keywords. irrationality; Lambert series; rational base; Hankel determinant; q -logarithm; Padé approximation; Lean 4. **MSC 2020.** 11J72 (primary); 11J82, 68V20 (secondary).

2 A rational base at which F is irrational

This section records the state of the argument on 2 September 2026. It is an ordinary complete proof consuming three published lemmas of Zudilin 2004, with the finite part Lean-checked and the whole not kernel-checked.

Write $F(t) = \sum_{m \geq 1} (t^m - 1)^{-1}$ as in the problem statement, and for coprime integers $a > b \geq 1$ put $t = a/b$. Two published thresholds bound the rational bases at which irrationality is currently reachable by a height argument. Duverney’s Théorème 2 [7, Th. 2, p. 174] covers $\log b / \log a < \frac{1}{3}(1 - 3/\pi^2) = 0.23207 \dots$, and Bundschuh and Väänänen’s

Theorem 2 in the case $\alpha = -1$ [6, Thm. 2, p. 177] covers the larger region

$$\frac{\log b}{\log a} < \frac{1}{2} - \frac{1}{\pi^2} = 0.398678 \dots$$

Zudilin's Theorem 1 [8, Thm. 1] gives the irrationality exponent $\mu = 2.46497868 \dots$ for the integer-base q -harmonic series in the direction $(\alpha_0, \alpha_1, \alpha_2; \beta) = (14, 12, 14; 27)$ with $m = 15$, and the closing paragraph of [9, §2] announces that the results extend to non-integer $p = r/s$ with $|p| > 1$ under an assumption $\log |r| > c \log |s|$ for some computable $c > 0$, without naming c and without printing a proof. The theorem below supplies that constant, namely $c = C_1/C_0 = \mu$, from the 2004 direction. It is an explicit-constant extension of an announced remark rather than a statement of a new kind, and what it buys is the strip $(0.398678, 0.405683]$, which contains $\log 4 / \log 31 = 0.403699$.

Theorem 2.1 (rational-base region). *Let $a > b \geq 1$ be coprime integers with*

$$b^\mu < a, \quad \text{equivalently} \quad \frac{\log b}{\log a} < \theta^* = \frac{C_0}{C_1} = 0.4056830213840605 \dots, \quad \mu = \frac{C_1}{C_0} = 2.4649786835749750 \dots$$

Then $F(a/b)$ is irrational. The proof consumes Lemma 7 of Zudilin [8] in its polynomial reading at display (24), together with his Lemmas 1 and 2 at the integer instance $p = 2$, and nothing else external.

In the normalisation of [9, §2], where the announced hypothesis is $\log |r| > c \log |s|$ for a computable $c > 0$, the constant is $c = \mu$, the irrationality exponent of Zudilin's own Theorem 1. Negative bases are not treated.

The two constants are the exponent budgets of Zudilin's direction. Explicitly

$$C_1 = (\alpha_0 + \alpha_1 + \alpha_2)\beta - \frac{1}{2}(\alpha_1^2 + \alpha_2^2 + \beta^2) = \frac{1091}{2}, \quad C_0 = 266 - \frac{3}{\pi^2}(225 - J) = 221.30008816500502 \dots,$$

where

$$J = \int_0^1 \omega(x) d(-\psi'(x)) = 77.94318447500909 \dots, \quad \omega(x) = \max\{0, [14x] + [13x] - [12x] - [15x], 2[14x] - [13x] - [15x]\}$$

so that $C_1/C_0 = \mu = 2.46497868 \dots$ and $\theta^* = 1/\mu$. Both published thresholds have this shape: Bundschuh–Väänänen's is $1/\mu_{BV}$ with $\mu_{BV} = 2\pi^2/(\pi^2 - 2) = 2.5082847619 \dots$, and $4^\mu = 30.4835 \dots < 31 < 4^{\mu_{BV}} = 32.3696 \dots$, which is exactly why $31/4$ sits between the two regions. The function ω takes the value 1 on exactly thirteen intervals, the first being $[1/14, 1/12)$ and the last $[13/14, 14/15)$; the list agrees with the one printed in [8].

Corollary 2.2. *$F((31/4)^r)$ is irrational for every integer $r \geq 1$.*

The strip $(1/\mu_{BV}, \theta^*]$ gained over Bundschuh–Väänänen has no member with $b = 2$ or $b = 3$. Its first member is $31/4$, followed by $53/5, 54/5, 56/5, 83/6, 85/6, 89/6$, and so on; the strip is infinite, and 137 coprime pairs with $a \leq 60$ lie in it. The problem's own base $3/2$ sits at $\theta = 0.630930 \dots$, far outside.

Theorem 2.3 (Archimedean cap on base-uniform rank-two families). *Let $(U_n, V_n) \in \mathbb{Z}[x]^2$ satisfy $\Lambda_n(x) = U_n(x)F(x) - V_n(x) \neq 0$, $\deg U_n, \deg V_n \leq \delta n^2(1 + o(1))$, and $\log |\Lambda_n(x)| = -\sigma n^2 \log x (1 + o(1))$ for every real $x > 1$, with $\sigma, \delta > 0$ independent of x . Then the homogenised forms $b^{\deg U_n} \Lambda_n(a/b)$ tend to zero exactly on $\log b / \log a < \sigma / (\sigma + \delta)$, and $\sigma / (\sigma + \delta) \leq 1/2$.*

So $b^2 < a$ is a hard cap on every base-uniform rank-two family in $\mathbb{Z}[x]$, Zudilin's included, and $3/2$ is out of reach for this construction family entirely. That is the reason the rest of this paper is a record of obstructions rather than a second theorem.

The height ratio is invariant under a common positive power, and

$$\frac{2}{5} < \frac{\log 4}{\log 31} < \frac{81}{200} < \theta^*.$$

The two rational comparisons are exact integer inequalities, $31^2 < 4^5$ and $4^{200} < 31^{81}$, and both are Lean-checked ([lower bound](#), [upper bound](#)), as is the invariance of the region under common powers ([power family](#)). The last comparison $81/200 < \theta^*$ is a finite rational computation: truncating each $\psi_1(u) - \psi_1(v)$ after nine terms gives $J > 389699/5000$, and $\pi > 157/50$ then gives $C_0 > 88371/400 = (81/200) \cdot (1091/2)$. The cutoff $81/200$ is an elementary rational sub-boundary of the contour computed here. It is not a threshold stated by Zudilin, and it should never be cited as one.

The base $31/4$ lies strictly outside the Bundschuh–Väänänen region, since $\log 4 / \log 31 > 2/5 > 1/2 - 1/\pi^2$; that exclusion is Lean-checked ([margin](#), [exclusion](#)). The base $3/2$ lies outside both regions, since $3^{81} < 2^{200}$ gives $81/200 < \log 2 / \log 3 = 0.6309\dots$ ([exclusion](#), [exclusion](#)). Nothing in this section bears on $t = 3/2$, and the rest of the paper is the record of why that base is hard.

What the published lemmas supply

Zudilin's Lemma 7 is a statement in $\mathbb{Z}[p]$. It gives

$$\Lambda_n(p) = p^{-M_n} D_N(p) \Omega_n(p)^{-1} F_n(p) = U_n(p) h_p(1) - V_n(p), \quad U_n, V_n \in \mathbb{Z}[p],$$

where $h_p(1) = \sum_{m \geq 1} (p^m - 1)^{-1}$, by Gaussian binomial integrality, cyclotomic clearing through the least common multiple, stability of the Heine transform, and Gauss's lemma for division by the monic Ω_n . The pair (U_n, V_n) is unique because $h_p(1) \notin \mathbb{Q}(p)$. The standing hypothesis $p \in \mathbb{Z}$ of [8] is used only later, at its equation (24), to pass from $\mathbb{Z}[p]$ to $\mathbb{Z}$; the argument here substitutes the rational base into the polynomial identity instead, and never enters that step. Lemma 2 is the equidistribution input. It gives

$$\frac{K_n - W_n}{n^2} \rightarrow 266 - \frac{3}{\pi^2} (225 - J) = C_0,$$

with weight $\psi_1(u) - \psi_1(v)$ over the intervals on which $\omega = 1$; only intervals with $u \geq 1/14$ occur, so no level $l > 15n$ contributes. Zudilin's Archimedean estimates (17) and (25) are not consumed, and no rational-base asymptotic theorem is needed or invoked.

Evidence boundary

Theorem 2.1 is an ordinary complete proof whose only external inputs are Lemmas 1, 2 and 7 of [8]. It is not kernel-checked. Lean checks the finite part: the trigamma definition of J , the bounds $81/200 < \theta^*$ from the single $k = 0$ term of each trigamma difference together with $\pi > 3.14$, and $\theta^* < 1/2$ from $J < \psi_1(1/14) < 198$; the membership of $31/4$ and of every power $(31/4)^r$ in the contour region; the exclusion of $3/2$ from it; and the exact integer bookkeeping of the forms. Those are in `ErdosProblems/Erdos1049/RationalBaseContour.lean` and `ErdosProblems/Erdos1049/ZudilinHeightRegion.lean`. The written proof is `ErdosProblems/Erdos1049/RationalBaseIrrationality_31_4_Proof.md`, and the contour derivation is `ErdosProblems/Erdos1049/ZudilinRationalHomogenisationDerivation.md`. The forms U_n, V_n were reconstructed exactly for $n \leq 3$ by an independent implementation, which verified the polynomial divisibility that Lemma 7 asserts, matched the closed-form degree bookkeeping to the computed degrees, confirmed nonvanishing of Λ_n and the Archimedean bracket on F_n , and reproduced the identity to relative error below 10^{-25} ; those receipts are the probes `formal_math/probes/erdos1049_zudilin_c0_c1_check.py` and `formal_math/probes/erdos1049_zudilin_exact_forms.py` and the logs under `state/formal_math/erdos257_period_noncollapse/erdos1049_contour_adversary/`. The convergence $(K_n - W_n)/n^2 \rightarrow C_0$ is exact at $n = 1, 2, 3$ and numerical at $n = 10, 100, 400$; the limit itself is Zudilin's Lemma 2, not a computation.

Priority is bounded rather than settled. Three rational-base regions are in print: Duverney 1996 at 0.23207 ..., Bundschuh–Väänänen 1994 at 0.398678 ..., and Zudilin 2016's unquantified announcement, whose constant is not named there. Zudilin 2004, Tachiya 2004, Bézivin 1988, Borwein 1991 and 1992, Van Assche 2001, Amdeberhan–Zeilberger 1998, and Postelmans–Van Assche 2006 all work at integer base. So does Matala-aho, Väänänen and Zudilin [10], whose standing hypothesis is $p = 1/q \in \mathbb{Z} \setminus \{0, \pm 1\}$ and which states on p. 880 that its methods do not sharpen Zudilin 2004. What is claimed here is the explicit constant and the resulting decidable membership test, not a new kind of theorem, and the theorem should be presented as proved rather than as first.

3 Exact normalized-Hankel order in Zudilin's construction

At $x = z = 1$, Zudilin's normalized moments are

$$v_m^* = \sum_{t \geq 0} q^{(m+1)t} \frac{(q; q)_m^3 (q^{t+1}; q)_m}{(q^{m+1+t}; q)_{m+1}}, \quad V_N^* = \det_{0 \leq i, j < N} (v_{i+j}^*).$$

His row transformation proves

$$\text{ord}_q V_N^* \geq \frac{N(N-1)(2N-1)}{6}$$

for every $N \geq 1$ [9, Section 4]. Tracking the first associated-graded coefficient shows that this estimate is always sharp.

Theorem 3.1 (sharp normalized Hankel order). *For every $N \geq 1$,*

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6},$$

and the coefficient of the first nonzero monomial is

$$[q^{N(N-1)(2N-1)/6}]V_N^* = \frac{(N!)^2(N+1)!}{2^N}.$$

Proof. We sharpen the source row operation before taking the determinant. For a formal series $H(X) = 1 + \sum_{r \geq 1} a_r X^r$, let $c_{j,t}$ be the coefficient of tail state zero after j applications of the associated-graded backward shift, starting in tail state t . A positive state moves down one place with coefficient -1 , while state zero emits state $s-1$ with coefficient a_s . Thus

$$c_{0,t} = \mathbf{1}_{t=0}, \quad c_{j+1,t} = -c_{j,t-1} \quad (t > 0), \quad c_{j+1,0} = \sum_{s \geq 1} a_s c_{j,s-1}.$$

If $h_r = [X^r]H(X)^{-1}$, the reciprocal identity $h_{r+1} = -\sum_{s=1}^{r+1} a_s h_{r+1-s}$ gives, by induction,

$$c_{j,t} = (-1)^j h_{j-t} \quad (0 \leq t \leq j), \quad c_{j,t} = 0 \quad (t > j).$$

For the actual normalized summand, the ratio of consecutive moment indices has associated-grade factor

$$H_t(X) = \frac{(1-X)^3(1-q^t X)^2}{(1-q^t X^2)(1-q^{t+1} X^2)}.$$

At $q = 0$ this is $(1-X)^3$ for $t \geq 1$, while $H_0(X) = (1-X)^4/(1+X)$. Consequently the $t \geq 1$ states contribute $(-1)^j \binom{j-t+2}{2}$, and state zero contributes

$$(-1)^j \frac{(j+1)(j+2)(2j+3)}{6}.$$

Summing $0 \leq t \leq j$ gives the transformed-row leading term

$$D_j v_{j+\ell}^* = (-1)^j \frac{(j+1)^2(j+2)}{2} q^{j(j+1)/2+j\ell} + O(q^{j(j+1)/2+j\ell+1}),$$

where D_j denotes the determinant-preserving source row operation.

After extracting the row factors, the first coefficient matrix is $(q^{j\ell})_{0 \leq j, \ell < N}$. Its Vandermonde determinant has order $N(N-1)(N-2)/6$; the row factors contribute $N(N-1)(N+1)/6$. Their sum is the asserted $N(N-1)(2N-1)/6$. The product of the row coefficients has absolute value

$$\prod_{j=0}^{N-1} \frac{(j+1)^2(j+2)}{2} = \frac{(N!)^2(N+1)!}{2^N},$$

and the row signs cancel the leading Vandermonde sign. Hence the first coefficient is positive and has the displayed value. $\square$

Consequence and evidence boundary. The theorem closes one named possibility: the normalized determinant cannot decay faster through an unrecorded increase in q -order. It does not extract a denominator factor, improve the remaining height balance, or prove irrationality at $3/2$. Lean checks the generic recurrence [at all depths](#) and its actual positive/zero tail specializations [and respectively](#). The exact tail fan-in [gives the row scalar](#), while the literal source recurrence holds for every index. The step is 1 below degree $n+1$ with first coefficient -5 or -3 and the exact operator recurrence holds on arbitrary sequences. The first transformed row is complete in every column: [packages exact order \$\ell+1\$ and leading coefficient \$-6\$](#) . The exact source D_j row matrix preserves every Hankel determinant [in power series](#); exact tails have unit first monomial [and moments isolate the zero tail below degree \$n+1\$ in every row](#). Lean checks determinant noncancellation at every rank, exact order and the factorial assembly [plus the generic unique-minimum bridge for filtered determinants](#). The serialized Lean replay and exact ranks 1 through 7 pass; only the filtered initial-monomial iteration for rows $j \geq 2$ and all columns remains formal.

4 The sharp 41/65 certificate

The numerical obstruction at $3/2$ is controlled by one small exact calculation. Writing it out is useful because the upper and lower inequalities play different roles: the upper inequality proves every strict analytic threshold below, while the lower inequality establishes the sharp rational scale. Exact failure of the rank-41 selector count one row earlier uses the additional integer comparison $2^{129} < 3^{82}$ below.

Theorem 4.1 (sharp power bracket). *One has*

$$2^{64} < 3^{41} < 2^{65}.$$

Consequently

$$\frac{41}{65} < \frac{\log 2}{\log 3}, \quad \frac{\log 3}{\log 2} < \frac{65}{41}.$$

Proof. Direct evaluation gives

$$\begin{aligned} 2^{64} &= 18446744073709551616, \\ 3^{41} &= 36472996377170786403, \\ 2^{65} &= 36893488147419103232. \end{aligned}$$

The logarithmic inequalities follow by taking logarithms and dividing by the positive numbers $41 \log 3$ and $41 \log 2$, respectively. $\square$

The integer sides are checked as [the upper certificate](#) and [the sharp lower certificate](#).

The upper half is stronger than the earlier $81/200$ comparison in the direction needed at $3/2$. Combining it with the elementary bound $1/2 - 1/\pi^2 < 2/5$ gives the following exact deficits.

For $\rho, \sigma \in \mathbb{R}$, write

$$\Theta_{\text{HP}}(\rho, \sigma) = \frac{(1 + \rho^2)/2 + \sigma - 3\sigma^2/\pi^2}{(1 + \rho)^2/2 + \sigma(1 + \rho) + (1 + \rho^2)/2 + \sigma}.$$

Corollary 4.2 (height and Hankel deficits). *For every $\rho, \sigma \in \mathbb{R}$ with $0 \leq \rho$ and $1 + \rho \leq \sigma$,*

$$\frac{3}{13} < \frac{\log 2}{\log 3} - \left(\frac{1}{2} - \frac{1}{\pi^2} \right), \quad \frac{3}{13} < \frac{\log 2}{\log 3} - \Theta_{\text{HP}}(\rho, \sigma),$$

where Θ_{HP} is the rectangular exponent threshold. Moreover

$$\frac{\log 3 / \log 2 - 1}{3} < \frac{8}{41}.$$

Proof. The first inequality follows from $41/65 - 2/5 = 3/13$, Theorem 4.1, and $1/2 - 1/\pi^2 < 2/5$. The rectangular threshold is no larger than the classical margin $1/2 - 1/\pi^2$, so the second follows. The final inequality is a direct rearrangement of $\log 3 / \log 2 < 65/41$. $\square$

The uniform rectangular bound and cubic charge bound are [checked here](#) and [checked here](#).

The next statement is the source-facing charge comparison. It does not prove the degree ceilings: those come from the scalar factor and the universally forced first-order southeast-border factor in the Zudilin model. It proves that even granting those ceilings, neither extraction reaches the required $39/41$ fraction of the raw charge.

Theorem 4.3 (scalar and border charge no-go). *For every integer $N > 0$,*

$$41(N^3 - N) < 39(4N^3 - 3N^2).$$

For every integer $N \geq 2$,

$$41(2N^3 - N) < 39(4N^3 - 3N^2).$$

Hence the same strict inequalities hold with the left side replaced by $41E$ whenever, respectively, $E \leq N^3 - N$ or $E \leq 2N^3 - N$.

Proof. After subtraction, the first inequality is

$$N(115N^2 - 117N + 41) > 0,$$

which holds for $N > 0$. The second becomes

$$N(74N^2 - 117N + 41) > 0.$$

For $N \geq 2$ the quadratic factor is positive and increasing. The assertions for E follow by monotonicity. $\square$

The application-facing downward-closed forms are [the scalar no-go](#) and [the scalar-plus-border no-go](#).

The formal source checks Theorem 4.1, Corollary 4.2, and Theorem 4.3 as exact Lean propositions. Their role is exclusion: scalar content and the forced first-order border do not supply enough charge. Higher residual valuations, determinant cancellation, a different integral model, and irrationality of $F(3/2)$ remain open.

5 Prime-supported rational gaps and two-row escape

A second strand concerns the arithmetic conclusion of a rational linear-form argument. Let $a, q, A, B \in \mathbb{Z}$ with $q > 0$, and write

$$\Lambda_{A,B}(a/q) = B \frac{a}{q} - A.$$

Here q is the denominator used to present the rational target, whereas B is a coefficient of an integral approximation row. They play different roles.

Theorem 5.1 (prime-supported two-row gap). *Let ℓ be prime and let $(A_1, B_1), (A_2, B_2) \in \mathbb{Z}^2$. Suppose*

$$\ell \nmid q, \quad \ell \mid B_1, \quad \ell \mid B_2, \quad \ell^2 \nmid A_1 B_2 - A_2 B_1.$$

Then

$$|\Lambda_{A_1, B_1}(a/q)| \geq \frac{1}{q} \quad \text{or} \quad |\Lambda_{A_2, B_2}(a/q)| \geq \frac{1}{q}.$$

The constant $1/q$ cannot be increased uniformly.

Proof. If ℓ divided both A_1 and A_2 , then ℓ^2 would divide both $A_1 B_2$ and $A_2 B_1$, contrary to the determinant hypothesis. Choose $i \in \{1, 2\}$ with $\ell \nmid A_i$. If $B_i a - A_i q = 0$, primality and $\ell \mid B_i$ would force $\ell \mid A_i$ or $\ell \mid q$, again a contradiction. Hence the integer $B_i a - A_i q$ is nonzero, and

$$|\Lambda_{A_i, B_i}(a/q)| = \frac{|B_i a - A_i q|}{q} \geq \frac{1}{q}.$$

For sharpness take $a = 1, q = 2, \ell = 3, (A_1, B_1) = (1, 3)$, and $(A_2, B_2) = (2, 3)$. The determinant is -3 , while both absolute remainders are $1/2$. $\square$

Corollary 5.2 (rational two-row no-double-decay). *Let $(A_{1,n}, B_{1,n})$ and $(A_{2,n}, B_{2,n})$ be sequences of integral rows satisfying*

$$A_{1,n} B_{2,n} - A_{2,n} B_{1,n} \neq 0 \quad \text{for every } n.$$

Then the two sequences

$$\Lambda_{A_{1,n}, B_{1,n}}(a/q), \quad \Lambda_{A_{2,n}, B_{2,n}}(a/q)$$

cannot both tend to zero.

Proof. At each n , the nonzero exterior determinant prevents both rational linear forms from vanishing. Every nonzero integral linear form at a/q has absolute value at least $1/q$. If both sequences tended to zero, then eventually both absolute values would be strictly smaller than $1/q$, a contradiction. $\square$

This corollary is an obstruction at rational targets, not an irrationality theorem. An application must still construct noncollinear rows whose two analytic remainders tend to zero under the contrary rationality hypothesis.

Proposition 5.3 (one-row prime-supported gap). *Let ℓ be prime, $\ell \mid B$, $\ell \nmid A$, and $\ell \nmid q$. Then*

$$|\Lambda_{A,B}(a/q)| \geq \frac{1}{q}.$$

The constant is sharp.

Proof. The numerator $Ba - Aq$ cannot vanish: otherwise $\ell \mid Aq$, and primality would contradict $\ell \nmid A$ and $\ell \nmid q$. Its absolute value is therefore at least one. Equality is attained by $a = 1, q = 2, \ell = 3, A = 1, B = 3$. $\square$

Corollary 5.4 (one-row prime-power-supported gap). *Let ℓ be prime, let $r \geq 1$, and suppose $\ell^r \mid B$, $\ell \nmid A$, and $\ell \nmid q$. Then*

$$|\Lambda_{A,B}(a/q)| \geq \frac{1}{q}.$$

Proof. Since $r \geq 1$, the prime ℓ divides B , and Proposition 5.3 applies. $\square$

The prime-power hypothesis records stronger denominator data but does not improve the sharp $1/q$ conclusion.

Proposition 5.5 (real unit-determinant height-decay tradeoff). *Let $F \in \mathbb{R}$, let $(A_1, B_1), (A_2, B_2) \in \mathbb{R}^2$, and let $u, v, w, z \in \mathbb{R}$ satisfy $|uz - vw| = 1$. Define the recombined rows by*

$$(A'_1, B'_1) = u(A_1, B_1) + v(A_2, B_2), \quad (A'_2, B'_2) = w(A_1, B_1) + z(A_2, B_2).$$

Suppose

$$|u|, |v|, |w|, |z| \leq H$$

and both recombined linear forms have absolute value at most $\varepsilon \geq 0$. Then

$$|A_1 B_2 - A_2 B_1| \leq 2H\varepsilon(|B_1| + |B_2|).$$

Proof. Writing $e_i = B'_i F - A'_i$, cancellation of the F -terms gives

$$A'_1 B'_2 - A'_2 B'_1 = e_2 B'_1 - e_1 B'_2.$$

Its absolute value is at most $\varepsilon(|B'_1| + |B'_2|)$. The determinant of the recombined rows is $(uz - vw)(A_1 B_2 - A_2 B_1)$, so its absolute value equals that of the original determinant. Finally each new denominator coordinate is at most $H(|B_1| + |B_2|)$, giving the displayed bound. $\square$

The transformation is real and has determinant of absolute value one; integer-unimodular changes of basis are a special case. Thus the proposition quantifies a genuine coefficient-height cost: a change of basis cannot make both remainders small without paying in the size of its coefficients.

Proposition 5.6 (zero-denominator selector collision). *Let $N \geq 1$ and $w_i \in (\mathbb{Z}/N\mathbb{Z})^2$ for $i < k$. If every second coordinate of w_i is zero and*

$$N < 2^k,$$

then two distinct binary selectors have the same vector sum.

Proof. Map a binary selector to the first coordinate of its sum. There are 2^k selectors and only N possible values, so two distinct selectors collide. Their second-coordinate sums are both zero by hypothesis. $\square$

The threshold is $N < 2^k$, rather than the ambient two-coordinate condition $N^2 < 2^k$, because the denominator coordinate has already collapsed.

Exact Lean identities. The six source-current declarations corresponding to the results above are, in theorem order:

```
rational_twoSelector_gap_of_prime_tail_support
rational_twoSelector_not_both_tendsto_zero
rational_integerLinearForm_gap_of_prime_support
rational_integerLinearForm_gap_of_prime_power_support
twoSelector_unimodular_height_decay_tradeoff
zmod_binary_collision_of_zero_denominator_coordinates
```

The formal statements have the same hypotheses and conclusions as the displayed results.

Exact remaining hypotheses. Nothing in this section proves that the actual q -Apéry rows contain a suitable prime in their denominator coordinates cofinally, proves the first-power determinant condition for a cofinal pair of rows, or supplies the analytic remainder estimates needed to invoke Corollary 5.2. These are the exact remaining hypotheses between the proved arithmetic consequences and an irrationality proof at $3/2$.

6 Integer scalar content is neutral

An irrationality argument by linear forms replaces the clearing scheme by explicit rational approximation: one constructs integer linear forms in 1 and $F(\beta)$ whose analytic decay outruns the height of their common denominator. For an integer coefficient pair (U, V) and a real target S , put

$$L_S(U, V) = US - V, \quad \Delta((U_n, V_n), (U_m, V_m)) = U_n V_m - U_m V_n.$$

We call $L_S(U, V)$ the *error* of the row at S ; a good approximation is one that makes it small. The second expression is the exterior determinant of the two rows, and it eliminates S exactly:

$$\Delta = U_m L_S(U_n, V_n) - U_n L_S(U_m, V_m).$$

This identity is what makes the determinant useful. Since Δ is an integer, if it does not vanish then

$$1 \leq |\Delta| \leq |U_m| |L_S(U_n, V_n)| + |U_n| |L_S(U_m, V_m)|,$$

and the two errors cannot both be smaller than $1/(|U_n| + |U_m|)$. The determinant therefore carries two competing quantities at once, an integer that divides it and its own absolute value, and the theorem below is about how a rescaling moves them.

Theorem 6.1 (integer-scalar-content no-go). *Let S be real, let (U_n, V_n) and (U_m, V_m) be pairs of integers, and let c_n, c_m be integers. Then*

$$L_S(c_n U_n, c_n V_n) = c_n L_S(U_n, V_n),$$

$$\Delta(c_n(U_n, V_n), c_m(U_m, V_m)) = c_n c_m \Delta((U_n, V_n), (U_m, V_m)),$$

and consequently

$$|\Delta(c_n(U_n, V_n), c_m(U_m, V_m))| = |c_n| |c_m| |\Delta((U_n, V_n), (U_m, V_m))|.$$

In particular $c_n c_m$ divides the scaled determinant. Hence a local divisor supplied only by the two integer scalar factors is paid for by exactly the same factor in the Archimedean determinant height.

Proof. All three identities are routine expansions in $\mathbb{Z}$ or $\mathbb{R}$; the divisibility statement uses the primitive determinant as its witness. $\square$

Informally, Theorem 6.1 says that multiplying specialised integer rows by scalar factors moves the local gain and the Archimedean cost by exactly the same amount. Within an argument whose only extra divisor is integer scalar content, primitive normalisation therefore loses no net gain. This says nothing about polynomial factors before specialisation, cross-row common factors, determinant-specific arithmetic, or additive combinations.

Example 6.2. Take $(U_n, V_n) = (1, 2)$ and $(U_m, V_m) = (3, 5)$, so that $\Delta = 1 \cdot 5 - 3 \cdot 2 = -1$. Multiplying the first row by $c_n = 6$ and the second by $c_m = 10$ gives the rows $(6, 12)$ and $(30, 50)$, whose determinant is $6 \cdot 50 - 30 \cdot 12 = -60$. That determinant is now divisible by 60, which looks like a local gain of 60; and its absolute value has risen from 1 to 60, which is a cost of exactly the same size.

Lean checks the error identity in [error scaling](#), the determinant identity in [content factorisation](#), the exact absolute-height identity in [absolute determinant scaling](#), and the divisor statement in [content-product divisibility](#). The elimination identity is the checked [exterior determinant identity](#).

The key point is that the two scalings are the same scaling. Each identity on its own is a one-line expansion, and the interest of the theorem is not in any one of them. Taken together they say that the factor $c_n c_m$ which a rescaling introduces into the determinant reappears undiminished, as $|c_n| |c_m|$, in the absolute value of that determinant. The third identity is displayed with absolute values for exactly that reason: it is what makes the statement one about the Archimedean height and not about divisibility alone. Whatever c_n and c_m are, a rescaling therefore leaves the balance between the local divisor and that height where it was.

The theorem does not construct primitive Padé rows, estimate their remainders, or prove that their exterior determinant is nonzero. It removes one source of apparent gain: multiplying a useful row by a large common integer cannot improve the local-to-Archimedean balance. Within a construction whose proposed gain comes only from those integer scalars, the rows may be primitive-normalised without a net loss. This

does not say that every candidate family must use such a normalisation or that every proof needs separate 2-adic and 3-adic gain. The theorem quantifies over arbitrary integer pairs, so it applies whenever a construction has reached that specialised-row stage.

7 Endpoint residues at (3, 2) and the four-jet kernel

Zudilin's treatment of q -harmonic series [8] builds linear forms of the shape used in Section 6 out of Heine's basic transform. The generic endpoint and jet lemmas below concern arbitrary integral coefficient pairs of this shape; they do not construct or instantiate Zudilin's actual polynomial family. Each lemma uses no property beyond integrality and quantifies over arbitrary elements of $\mathbb{Z}[X]$. The exception is Theorem 7.12 at the end of the section, which concerns the scalar parameters of Zudilin's cone rather than the coefficient polynomials.

Substituting $X = 3/2$ into an integer polynomial produces a rational number, and multiplying by a power of 2 clears its denominator. The following evaluation records that cleared numerator, so that all the arithmetic below stays inside $\mathbb{Z}$. For $P(X) = \sum_i p_i X^i \in \mathbb{Z}[X]$ and a declared width $W \geq 0$, put

$$H_W(P) = \sum_{i=0}^W p_i 3^i 2^{W-i}.$$

This is the [homogeneous endpoint evaluation](#). It is homogeneous in the sense that X^i is replaced by $3^i 2^{W-i}$, so the numerator and the denominator of the base are carried symmetrically. When $W \geq \deg P$ it is exactly the cleared numerator, since

$$\sum_{i=0}^W p_i 3^i 2^{W-i} = 2^W \sum_{i=0}^W p_i \left(\frac{3}{2}\right)^i = 2^W P\left(\frac{3}{2}\right).$$

Theorem 7.1 (endpoint residues). *Let $P = \sum_i p_i X^i \in \mathbb{Z}[X]$ and let $W \geq 0$. Then*

$$H_W(P) \equiv p_0 2^W \pmod{3}, \quad H_W(P) \equiv p_W 3^W \pmod{2}.$$

Consequently a unit constant coefficient prevents divisibility by 3, and a unit coefficient at the declared width W prevents divisibility by 2.

Proof. Modulo 3, every summand with $i > 0$ vanishes; modulo 2, every summand with $i < W$ vanishes. The remaining powers are units in the corresponding residue fields. $\square$

Since 2^W is invertible modulo 3 and 3^W is invertible modulo 2, the two congruences say more than the stated consequence: divisibility of $H_W(P)$ by 3 is decided by p_0 alone, and divisibility by 2 by p_W alone. The rest of the coefficient vector is invisible to both primes. The coefficient p_W is the top coefficient of P exactly when $\deg P = W$, and is zero when $\deg P < W$; the identity $H_W(P) = 2^W P(3/2)$ likewise holds only for $\deg P \leq W$.

Example 7.2. Take $W = 2$. The three polynomials below differ only at an endpoint.

P	$H_2(P)$	$3 \mid H_2(P)$	$2 \mid H_2(P)$
$X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 13$	no	no
$X^2 + 3$	$3 \cdot 4 + 0 \cdot 6 + 1 \cdot 9 = 21$	yes	no
$2X^2 + 1$	$1 \cdot 4 + 0 \cdot 6 + 2 \cdot 9 = 22$	no	yes

The first has both endpoints equal to 1 and its evaluation, 13, is divisible by neither prime; as a check, $2^2((3/2)^2 + 1) = 13$. The second and third show that each hypothesis is used: spoiling the constant endpoint admits 3, and spoiling the top endpoint admits 2.

The congruences are the [bottom-endpoint identity](#) and [top-endpoint identity](#); the unit consequences are the [constant-endpoint obstruction](#) and [top-endpoint obstruction](#).

The proof is two lines, but the shape of the statement is not incidental. The difficulty lies in the fact that the specialisation sees a different endpoint at each of the two primes: modulo 3 only the constant coefficient survives, and modulo 2 only the top one does. The two exclusions are therefore conditions at opposite ends of the coefficient vector, and the statement below imposes one at each end, on the two entries of a single coefficient pair.

Proposition 7.3 (common divisor). *Let $U, V \in \mathbb{Z}[X]$ and let $W \geq 0$. If U has unit top endpoint, V has unit constant endpoint, and an integer c divides both $H_W(U)$ and $H_W(V)$, then*

$$2 \nmid c \quad \text{and} \quad 3 \nmid c.$$

This is the checked [common-divisor exclusion](#). The proposition does not say that the two evaluations are coprime; it says that whatever they share misses both of the primes that matter at $3/2$.

Example 7.4. Take $W = 2$, $U = X^2 + 3$ and $V = 5X^2 + 1$. The top endpoint of U and the constant endpoint of V are both 1, and

$$H_2(U) = 3 \cdot 4 + 1 \cdot 9 = 21, \quad H_2(V) = 1 \cdot 4 + 5 \cdot 9 = 49.$$

Here $\gcd(21, 49) = 7$, so a common divisor does exist and is not small; it is simply coprime to 6.

Corollary 7.5 (no gain from integer scalar content or the stated common divisor at $3/2$). *Under the endpoint hypotheses of Proposition 7.3, neither integer scalar content of specialised rows nor a common divisor of the two specialised evaluations $H_W(U)$ and $H_W(V)$ can supply factors 2 and 3 by those mechanisms in the common-width endpoint architecture studied here.*

Proof. By Theorem 6.1 integer scalar content multiplies the analytic error and the exterior determinant, including the absolute determinant height, by exactly the factors it introduces, so a divisor obtained that way is paid for by the same factor in that height. By Proposition 7.3 an integer dividing both specialised evaluations is divisible by neither 2 nor 3. $\square$

One further consequence of Theorem 7.1 is worth stating, because it bears on the most natural way one might hope to import an existing denominator reduction. Write Φ_m for the m th cyclotomic polynomial and, for coprime $a > b \geq 1$, put $\Phi_m(a, b) = b^{\varphi(m)} \Phi_m(a/b)$ for its homogenisation at the declared width $\varphi(m) = \deg \Phi_m$.

Proposition 7.6 (homogenised cyclotomic values are unit at both endpoints). *Let $a > b \geq 1$ with $\gcd(a, b) = 1$ and let $m \geq 1$. Then $\gcd(\Phi_m(a, b), ab) = 1$. In particular $\gcd(\Phi_m(3, 2), 6) = 1$ for every m .*

Proof. Φ_m is monic and $\Phi_m(0) = \pm 1$, so its coefficients at both declared endpoints are units. Theorem 7.1 applies verbatim with W replaced by $\varphi(m)$: modulo a prime dividing b only the top term of the homogenisation survives, and modulo a prime dividing a only the constant term does. $\square$

The kernel-checked declaration `cyclotomic hom eval is coprime mul` proves Proposition 7.6 in the same homogeneous-evaluation representation. It checks Proposition 3.6 under the displayed coprimality assumptions; it does not certify the later analytic deductions or Proposition 8.6.

The methods that reduce denominators at integer bases, the factorial-coset quotients of Rhin and Viola [13] and the order-twelve group and cyclotomic divisor of Zudilin [8], produce their gain as cyclotomic or factorial factors of the coefficient polynomials. Proposition 7.6 says that transporting such a factor through the homogenisation at $(3, 2)$ contributes no power of 2 and no power of 3, whatever its size. This does not make such factors useless: a large odd divisor still reduces Archimedean height, and that is a different account of the same product formula. It does say that the 2- and 3-primary gain the architecture of Section 12 requires must come from somewhere other than an imported cyclotomic factor. The exclusions therefore reflect the architecture itself, beyond the local behaviour of the two primes involved.

Both ingredients are Lean-checked; the combination is an ordinary deduction and is not separately formalised. The corollary excludes two ways of producing the targeted gain. It does not show that a gain of that kind is necessary for a proof by linear forms at $3/2$.

Corollary 7.5 excludes the two multiplicative mechanisms above, and one candidate pursued in the rest of this section is additive: rather than multiplying one row by a scalar, take an integer combination of several rows and ask that the combination be divisible where the individual rows are not. The endpoint congruences are the first case of a divisibility condition that can be imposed to any depth, and it is that condition, read additively, which is counted below.

We first raise the two congruences to prime powers. Fix depths $R, S \geq 0$. For $P \in \mathbb{Z}[X]$ the *bottom jet* $J_{3,R}(P)$ is the residue of $H_W(P)$ modulo 3^R , and the *top jet* $J_{2,S}(P)$ is its residue modulo 2^S ; Theorem 7.1 computes them at $R = S = 1$. Their vanishing is exactly the requested divisibility:

$$J_{3,R}(P) = 0 \iff 3^R \mid H_W(P), \quad J_{2,S}(P) = 0 \iff 2^S \mid H_W(P).$$

These are the checked `bottom-jet divisibility criterion` and `top-jet divisibility criterion`.

The *four-jet signature* of a coefficient pair (U, V) is then the quadruple

$$(J_{3,R}(U), J_{3,R}(V), J_{2,S}(U), J_{2,S}(V)) \in (\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

two residues for each of the two primes, one from each entry of the pair. By the displayed criteria it vanishes exactly when 3^R divides both specialised entries and 2^S divides both.

For this candidate architecture, this turns the targeted local divisor into an additive congruence-kernel problem: what is sought is no longer a common divisor of the two evaluations, but a vector of small integer coefficients on which four residues vanish at once. Since H_W is linear in the coefficients of P , the four-jet signature of a combination is the corresponding combination of signatures, which is what makes the following count possible.

Theorem 7.7 (binary four-jet collision). *Fix a width W and depths R, S , and let $(U_j, V_j)_{j < M}$ be any M pairs of integral polynomials. Call a subset of $\{0, \dots, M-1\}$, equivalently a vector of $\{0, 1\}^M$, a binary selector. If the 2^M binary selectors outnumber the finite four-jet target*

$$(\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

then two distinct subsets have the same four-jet sum. Subtracting their indicator vectors gives a nonzero coefficient vector in $\{-1, 0, 1\}^M$ cancelling all four jets. The target has exact cardinality

$$(3^R)^2(2^S)^2.$$

In particular, if $R > 0$ and $4R + 2S \leq M$, such a collision exists.

Proof. Send each binary selector to the sum of the four-jet signatures it selects. The claimed cardinal inequality and the pigeonhole principle give two distinct selectors in the same fibre. The cardinality formula is the product of the four cyclic-modulus cardinalities. For $R > 0$,

$$(3^R)^2(2^S)^2 < (4^R)^2(2^S)^2 = 2^{4R+2S} \leq 2^M,$$

which proves the stated sufficient threshold. □

The power bracket improves the generic coefficient $4R$ when the bottom depth is a multiple of 41.

Corollary 7.8 (sharp rank-41 four-jet threshold). *Let $T > 0$. At bottom depth $R = 41T$, any family of $M \geq 130T + 2S$ integral polynomial pairs has two distinct binary selectors with the same four-jet sum. For $T = 1$ the coefficient 130 is exact for this counting argument:*

$$2^{129+2S} < |(\mathbb{Z}/3^{41}\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2|.$$

No exact-optimality assertion is made here for $T > 1$.

Proof. The upper power inequality gives

$$(3^{41T})^2(2^S)^2 < (2^{65})^{2T}(2^S)^2 = 2^{130T+2S} \leq 2^M,$$

so Theorem 7.7 applies. For $T = 1$, direct integer evaluation gives $2^{129} < 3^{82}$; multiplying by $(2^S)^2$ gives the displayed reverse count at $129 + 2S$. □

The sufficient collision threshold is [checked here](#), while its exact unit-block failure one row earlier is [checked here](#).

Counting alone does not ensure that the two selectors produce different analytic remainders. The exact missing step is a bounded-fibre estimate.

Theorem 7.9 (bounded-fibre escape). *Let A and B be finite sets, let $f : A \rightarrow B$, and let $g : A \rightarrow C$ be any map into a set C . Suppose every fibre of g has at most k elements. If*

$$|B|k < |A|,$$

then there exist distinct $x, y \in A$ such that

$$f(x) = f(y) \quad \text{and} \quad g(x) \neq g(y).$$

Thus, with f the four-jet sum and g the analytic remainder, a uniform remainder-multiplicity bound converts surplus selector entropy into a four-jet collision outside the remainder nullspace.

Proof. If every pair in a common f -fibre also had the same g -value, each f -fibre would lie in one g -fibre and hence have size at most k . Summing over the at most $|B|$ fibres of f would give $|A| \leq |B|k$, contrary to the hypothesis. $\square$

This finite escape principle is [checked here](#).

Theorem 7.10 (Bézout–Plücker tail collapse). *Let R_0 be a commutative ring and let $w_n = (A_n, B_n) \in R_0^2$. Suppose that every B_n is a unit and every adjacent minor vanishes:*

$$A_n B_{n+1} - B_n A_{n+1} = 0 \quad (n \geq 0).$$

Then every pairwise minor $A_i B_j - B_i A_j$ vanishes. In particular, take $R_0 = \mathbb{Z}/(2^S 3^R)\mathbb{Z}$ with $R > 0$. If $S + 2R \leq k$, there are two distinct binary selectors $s, t \in \{0, 1\}^k$ such that

$$\sum_{i < k} s_i w_i = \sum_{i < k} t_i w_i.$$

Thus the sufficient width is $S + 2R$, rather than the ambient two-coordinate width $2S + 4R$.

Proof. A unit second coordinate makes each row a unimodular anchor. Vanishing of the next minor therefore writes the next row as a scalar multiple of the current one; induction places the entire tail on the line through w_0 and proves the pairwise-minor assertion. A determinant-one Bézout shear sends w_0 to $(1, 0)$, so all selector sums have only one free residue coordinate and occupy at most $2^S 3^R$ values. Finally

$$2^S 3^R < 2^S 4^R = 2^{S+2R} \leq 2^k,$$

and pigeonhole gives the two selectors. $\square$

The minor propagation is [kernel checked](#); the one-dimensional collision after a Bézout shear is [kernel checked](#); and the explicit 2/3-depth endpoint is [kernel checked](#). This conditional theorem is stronger than the ambient four-jet count only after its minor-vanishing hypothesis has been established. No such all-tail hypothesis is proved here

for an actual q -Apéry or Zudilin family, and the theorem says nothing about whether the resulting selector difference has nonzero analytic remainder.

Boundary. Corollary 7.8 is a sharp finite kernel statement at $T = 1$, not an analytic nonvanishing theorem. Theorem 7.9 identifies the precise extra input needed to escape the nullspace, but this paper does not prove a multiplicity bound for the actual q -Apéry or Zudilin remainder family.

The target count is the checked [four-jet target cardinality](#); the abstract collision is the checked [four-jet pigeonhole kernel](#), and the linear sufficient condition is the checked [rank–depth collision threshold](#). The count is a routine pigeonhole; the reformulation above is what makes it relevant. Pigeonhole cancellation itself requires no independence. Additional information about the input family is needed to ensure that the resulting nonzero selector difference has a nonzero combined polynomial pair and analytic remainder. None of the statements proved here supplies such a family or proves either nonvanishing conclusion.

Example 7.11. At depths $R = S = 1$ the four-jet target is $(\mathbb{Z}/3\mathbb{Z})^2 \times (\mathbb{Z}/2\mathbb{Z})^2$, of cardinality $9 \cdot 4 = 36$, and the threshold reads $M \geq 4 \cdot 1 + 2 \cdot 1 = 6$. With six pairs there are $2^6 = 64$ binary selectors against 36 targets, so two of them collide and their difference is a vector in $\{-1, 0, 1\}^6$, not identically zero, killing all four jets.

Informally, the theorem says only this: once there are at least $4R + 2S$ pairs and the bottom depth R is positive, some coefficient vector in $\{-1, 0, 1\}^M$, not identically zero, kills all four jets of the corresponding combination. It does not say that the combination is nonzero as a pair of polynomials, and it does not say that its remainder is nonzero. Those are the two obligations Problem 12.1 carries.

One further exclusion is recorded here. It concerns the scalar parameters of Zudilin’s cone rather than the coefficient polynomials.

Theorem 7.12 (scalar margin no-go). *Let $C_1 > 0$. If $C_0 \leq 0$ or $2C_0 \leq C_1$, then*

$$C_0 \log 3 - C_1 \log 2 < 0.$$

In the positive branch $C_0 > 0$ and $2C_0 \leq C_1$, the stronger estimate is

$$C_0 \log 3 - C_1 \log 2 < -\frac{17}{41} C_0 \log 2.$$

Written multiplicatively, the conclusion is $3^{C_0} < 2^{C_1}$. The inequality is immediate from $\log 3 < 2 \log 2$ and $C_1 > 0$, and is the checked [three-halves scalar margin](#). The positive-branch deficit is the checked [17/41 margin](#). The interest is in the fence around it. The primary Zudilin theorem [8, Theorem 1, p. 154; Sec. 5, pp. 161–162] supplies an integer-base irrationality-exponent estimate on its parameter cone, and the elementary inequality $\mu \geq 2$ then forces $2C_0 \leq C_1$ whenever $C_0 > 0$; Lean checks that implication separately. The primary theorem assumes an integer base $p = 1/q$. It does not state a rational $p = 3/2$ theorem, so the all-scale coefficient construction and rational specialisation remain external to the checked result, which is the scalar parameter margin alone.

8 Failure of coordinatewise clearing at $3/2$

This section and the next return to the elementary route and record what the residue s^n costs there. We first isolate the arithmetic that the clearing scheme leaves behind, in a form that does not mention the series. The point of doing so is that the clearing argument asks for two things at once, that a power of the numerator divide what has been accumulated and that what survives be small, and these are easier to play off against each other once the series has been discarded and only six natural numbers remain.

Definition 8.1 (coordinatewise corridor). Let a, b, N, K, Q, D be natural numbers. Say that (a, b, N, K, Q, D) is a *coordinatewise corridor* when

$$a > 0, \quad Q > 0, \quad D > 0, \quad D \leq N + K, \quad a^K \mid QD, \quad Qb^{N+K+1} < a^{K+1}.$$

The reading is: a and b are the numerator and denominator of the base, playing the roles of r and s in the introduction, so that $(a, b) = (3, 2)$ is the case of interest; N is the shift, K is the width of the cleared window, Q is the accumulated clearing factor, and D is the final coefficient being cleared. The bound $D \leq N + K$ is the only property of the coefficient used; for the divisor-counting coefficient it holds because $\tau(n) \leq n$. The divisibility $a^K \mid QD$ is the requirement that clearing succeeded coordinatewise, and the last inequality is the tail estimate that makes the trapped integer smaller than 1. The name records the shape of the constraint: the divisibility bounds a^K from above by $Q(N + K)$, the tail estimate bounds a^{K+1} from below by Qb^{N+K+1} , and admissible parameters must fit in the band between them. The definition is the [corridor predicate](#).

Theorem 8.2 (corridor bound). If (a, b, N, K, Q, D) is a *coordinatewise corridor*, then

$$b^{N+K+1} < a(N + K).$$

Proof. A routine divisibility computation. Since $QD > 0$ and $a^K \mid QD$, we have $a^K \leq QD$, and $D \leq N + K$ gives $a^K \leq Q(N + K)$. Hence

$$Qb^{N+K+1} < a^{K+1} = a^K \cdot a \leq Q(N + K) \cdot a,$$

and cancelling the positive factor Q gives the claim. $\square$

Formalised as the [power-versus-linear consequence](#).

The inequality of Theorem 8.2 is where the integer and rational cases part. At $b = 1$ it reads $1 < a(N + K)$, which holds for every $a \geq 2$ and every nonempty window; this necessary inequality imposes no obstruction. The other corridor hypotheses remain in force. At $b \geq 2$ the left side is exponential in $N + K$ and the right side is linear, so the corridor can survive only for small $N + K$. At $(a, b) = (3, 2)$ the crossing has already happened at the smallest admissible window.

Example 8.3. Take the smallest window, $N = K = 1$, and numerator $a = 3$. At $b = 1$ the tuple $(3, 1, 1, 1, 3, 1)$ is a corridor: $D = 1 \leq 2$, the divisibility reads $3 \mid 3$, and the tail estimate reads $3 \cdot 1^3 = 3 < 3^2 = 9$. At $b = 2$ no choice works. The tail estimate becomes $Q \cdot 2^3 < 3^2$, which forces $Q = 1$; the divisibility then reads $3 \mid D$, and the only candidates are $D = 1$ and $D = 2$, neither divisible by 3. This is the proof of Theorem 8.2 in miniature: it derives $a^K \leq Q(N + K)$, here $3 \leq 2$.

Proposition 8.4. *For every natural number $x \geq 2$ we have $3x < 2^{x+1}$.*

Proof. A routine induction from $x = 2$, where $6 < 8$. For the step, $2^{x+1} \geq 2^2 > 3$ when $x \geq 1$, so $3(x+1) = 3x+3 < 2^{x+1} + 2^{x+1} = 2^{x+2}$. $\square$

Formalised as the [exponential comparison](#).

Theorem 8.5 (no corridor at base $3/2$). *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D , the tuple $(3, 2, N, K, Q, D)$ is not a coordinatewise corridor.*

Proof. A corridor would give $2^{N+K+1} < 3(N+K)$ by Theorem 8.2, contradicting Proposition 8.4 applied to $x = N+K \geq 2$. $\square$

Formalised as the [corridor exclusion at three halves](#).

Theorem 8.5 excludes the coordinatewise clearing scheme at $3/2$, and nothing else: it does not bound the denominator of $F(3/2)$, it does not show that $F(3/2)$ is irrational, and it does not show that $F(3/2)$ is rational. It also does not cover a clearing scheme of a different shape, since the corridor fixes one divisibility pattern and one tail inequality.

9 The cleared-tail recurrence and the size of the forcing term

Theorem 8.5 says that one scheme fails. This section identifies the quantity responsible, as an exact recurrence.

Let r, s, B, F be rationals with $r \neq 0$ and let $c : \mathbb{N} \rightarrow \mathbb{Q}$ be arbitrary. Define the prefix and the cleared tail state by

$$P_N = \sum_{m=0}^{N-1} \frac{c(m+1) s^{m+1}}{r^{m+1}}, \quad U_N = B r^N (F - P_N). \quad (*)$$

Thus P_N is the partial sum of $\sum_{n \geq 1} c(n)(s/r)^n$ through level N , and U_N is the tail of a putative value F after that level, scaled by $B r^N$. These are the [rational-base prefix](#) and the [cleared tail state](#).

Theorem 9.1 (cleared-tail recurrence). *Let $r, s, B, F \in \mathbb{Q}$ with $r \neq 0$, let $c : \mathbb{N} \rightarrow \mathbb{Q}$, and let P_N and U_N be as in $(*)$. Then for every N ,*

$$U_{N+1} = r U_N - B c(N+1) s^{N+1}.$$

Proof. An immediate computation. Expanding $P_{N+1} = P_N + c(N+1)s^{N+1}/r^{N+1}$ and $r^{N+1} = r^N \cdot r$ in the definition of U_{N+1} and clearing the denominator r^{N+1} , which is nonzero, gives the identity. $\square$

Formalised as the [cleared-tail recurrence](#).

The recurrence has a linear part $r U_N$ and a forcing term $B c(N+1) s^{N+1}$, the inhomogeneous term the state receives at each step. The direct-clearing route studied here works only if the state remains in a bounded window, and the forcing term is what that window must absorb. Its size is what separates the two denominator regimes.

Theorem 9.2 (the forcing term). *Let s, B be natural numbers and $c : \mathbb{N} \rightarrow \mathbb{N}$, and put $G_N = B c(N+1) s^{N+1}$.*

1. If $s \geq 2$, $B \geq 1$ and $c(N+1) \geq 1$, then $2^{N+1} \leq G_N$.
2. If $s = 1$, then $G_N = Bc(N+1)$.

Proof. Both parts are routine. For the first, $2^{N+1} \leq s^{N+1} = 1 \cdot s^{N+1} \leq Bc(N+1)s^{N+1}$, using $Bc(N+1) \geq 1$. The second part is the definition with $s = 1$. $\square$

Formalised as the [exponential lower bound](#) and the [integer-base collapse](#).

At $s = 1$ the forcing term is $Bc(N+1)$, so it grows only as fast as the coefficient; for the divisor function that is $O(N^\varepsilon)$ for every $\varepsilon > 0$, and a bounded-state argument has room. At $s \geq 2$ the same term is at least 2^{N+1} whenever the coefficient is nonzero.

Example 9.3. Take $B = 1$, $c = \tau$ and $N = 9$, so that the coefficient is $\tau(10) = 4$. At $s = 1$ the forcing term is 4. At $s = 2$ it is $4 \cdot 2^{10} = 4096$, and part (1) of Theorem 9.2 already guarantees at least $2^{10} = 1024$ without knowing the coefficient at all.

This is an exact lower bound on one quantity, and it is all that is proved. It is not a proof that no bounded-state argument exists at $s \geq 2$; the theorem that one particular scheme fails is Theorem 8.5, and the bound here records the size of the term that scheme would have to absorb.

10 The height criterion at 7/2

In 1994 Bundschuh and Väänänen proved an irrationality criterion for a family of rational bases cut out by a height condition [6, Theorem 2, p. 177; hypotheses pp. 175–176]. We keep their notation: q is the base, and α and λ are the parameters of the criterion. In its special case $\alpha = -1$, the printed hypothesis is

$$\lambda < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

At $q = 7/2$ the Archimedean parameter is $\lambda = \log 7 / \log(7/2)$. The criterion therefore applies once the following strict inequality is checked.

Theorem 10.1 (the 7/2 height condition).

$$\frac{\log 7}{\log(7/2)} < \left(\frac{1}{2} + \frac{1}{\pi^2} \right)^{-1}.$$

Numerically the two sides are 1.5533 ... and 1.6630 ..., so the condition holds with a margin of about 0.11. The Lean proof factors the estimate through the explicit [height-region predicate](#) and the [integer certificate](#) $2^{18} < 7^7$, the resulting [logarithmic ratio bound](#) $\log 2 / \log 7 < 7/18$, the [\$\pi\$ -bound](#) $1/\pi^2 < 1/9$, and the [strict margin](#)

$$\frac{\log 2}{\log 7} < \frac{1}{2} - \frac{1}{\pi^2}.$$

The [final height inequality](#) then rewrites $\log(7/2) = \log 7 - \log 2$ and closes the displayed condition.

This formalises the complete elementary parameter check at $q = 7/2$; it does not formalise Bundschuh and Väänänen’s analytic irrationality theorem, whose proof occupies pp. 189–193 of the source. The conclusion that $F(7/2)$ is irrational is consequently cited from that theorem, not claimed as a Lean theorem here.

The criterion does not cover $3/2$, and the results of Sections 8 and 9 give no evidence either way about whether $F(3/2)$ is irrational.

10.1 The 81/200 logarithmic region

Consider the rational-height region

$$\frac{\log b}{\log a} < \frac{81}{200} \quad (a > b > 0).$$

No bibliographic source for an analytic irrationality theorem at this cutoff is asserted here. The Lean module instead treats the displayed inequality as a definition and proves elementary memberships and exclusions. In particular,

$$\frac{2}{5} < \frac{\log 4}{\log 31} < \frac{81}{200} < \frac{\log 2}{\log 3}.$$

The first two comparisons come respectively from $31^2 < 4^5$ and $4^{200} < 31^{81}$; the last comes from $3^{81} < 2^{200}$. The lower bound is the checked theorem [two-fifths lower bound](#). Consequently $31/4$, and every positive power $(31/4)^r$, lies in the enlarged region ([checked](#), [power family](#)). The same base lies strictly outside the earlier Bundschuh–Väänänen region ([checked](#)), so the 81/200 inequality defines a strict set-theoretic enlargement of the earlier recorded logarithmic region. Membership alone supplies no irrationality theorem; the theorem for $31/4$ comes from Section 2, which places 81/200 below the homogenisation threshold θ^* and imports two lemmas of Zudilin.

It still does not approach $3/2$. The exact comparison

$$3^{81} < 2^{200} \quad \Rightarrow \quad \frac{81}{200} < \frac{\log 2}{\log 3}$$

is Lean-checked, as is the conclusion that $3/2$ belongs to neither height region ([boundary](#), [exclusion](#)). Thus 81/200 is the larger of the two explicitly defined cutoffs used below, while a cutoff that includes $3/2$ must be strictly larger than $\log 2 / \log 3 \approx 0.6309$.

11 Denominator exponents for a homogenised Padé construction

A second external route builds the linear forms of Section 6 by Padé approximation. Such a construction produces its coefficients as sums of rational summands, and to obtain integer rows one multiplies through by a single common denominator. The book-keeping obligation is then to check that no summand needs a larger denominator than the one proposed, which is a comparison between the exponents alone. This section does that comparison, and only that.

No source construction is named here for the exponent expressions below, and they are checked as displayed polynomial identities rather than derived. Giving them Padé-theoretic force would additionally require deriving the expressions from a stated homogenised coefficient formula, proving integrality of the coefficients after multiplication by the proposed common denominator, and establishing nonvanishing and decay of the associated remainder. None of those three is claimed here.

For a homogenised construction over integer parameters the proposed common denominator exponent is $E_n = (3n^2 - n)/2$. Since only doubled exponents occur below, every statement lives over $\mathbb{Z}$ and no parity bookkeeping is needed; we write $\tilde{E}_n = 2E_n = 3n^2 - n$.

Proposition 11.1 (summand bound and exact gap). *Let $\tilde{E}_n = 3n^2 - n$ and put*

$$\tilde{P}(n, k) = 2(k(n - k) + nk) + k(k - 1),$$

$$\tilde{Q}(n, m) = 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \quad j = n - m - 1.$$

Then, for integers n, k, m :

1. if $0 \leq k \leq n$, then $\tilde{P}(n, k) \leq \tilde{E}_n$, and the gap factors as $\tilde{E}_n - \tilde{P}(n, k) = (n - k)(3n - k - 1)$;
2. $\tilde{E}_n - \tilde{Q}(n, m) = 2(n + m(m - 1))$ identically;
3. if $n \geq 0$ and $m \geq 1$, then $\tilde{Q}(n, m) \leq \tilde{E}_n$.

Part (1) is the [summand exponent bound](#), part (2) the [exact gap identity](#), and part (3) the [maximal exponent bound](#). The gap in (2) is an identity in $\mathbb{Z}[n, m]$, so (3) follows from $m(m - 1) \geq 0$ and $n \geq 0$; the hypothesis $m \geq 1$ names the range in which the corresponding summand does not vanish, and it is not the sharpest hypothesis under which the inequality holds.

Example 11.2. At $n = 2$ the proposed doubled exponent is $\tilde{E}_2 = 10$, and $\tilde{P}(2, k)$ takes the values 0, 6, 10 at $k = 0, 1, 2$. The three gaps are 10, 4, 0, matching the factorisation $(2 - k)(5 - k)$ of part (1); the summand at $k = 2$ is the one that saturates the proposed denominator. For part (2), at $m = 1$ we have $j = 0$ and $\tilde{Q}(2, 1) = 6$, with gap $4 = 2(2 + 1 \cdot 0)$.

These are routine inequalities between polynomials in the exponents. They establish that the proposed exponent $\tilde{E}_n$ dominates the two displayed summand exponent expressions, and nothing further. Positivity of the remainder, its rate of decay, and the comparison of that rate against the denominator height are the analytic obligations, and none of them is treated here, so nothing in this section is an irrationality measure.

12 Complements and further questions

Problem #1049 remains open. Theorem 7.7 suggests the following precise sufficient subproblem for one common-width additive architecture. It is not asserted to be necessary for every proof of irrationality.

Problem 12.1 (common-width simultaneous endpoint-jet construction). Exhibit an integer constant $C \geq 1$ and, for every sufficiently large positive integer n , positive integers W_n, R_n, S_n, M_n such that

$$n^2 \leq W_n, R_n, S_n \leq Cn^2, \quad 4R_n + 2S_n \leq M_n \leq Cn^2,$$

together with polynomial pairs $(U_{n,j}, V_{n,j}) \in \mathbb{Z}[X]^2$ for $0 \leq j < M_n$, each of degree at most the common declared width W_n , whose specialised integer rows are primitive:

$$\gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j})) = 1.$$

Find a nonzero vector $\lambda^{(n)} \in \{-1, 0, 1\}^{M_n}$ for which, on putting

$$U_n = \sum_{j < M_n} \lambda_j^{(n)} U_{n,j}, \quad V_n = \sum_{j < M_n} \lambda_j^{(n)} V_{n,j},$$

the pair (U_n, V_n) is not $(0, 0)$, all four common-width jets vanish,

$$J_{3,R_n}(U_n) = J_{3,R_n}(V_n) = 0, \quad J_{2,S_n}(U_n) = J_{2,S_n}(V_n) = 0,$$

where every jet in this display is formed using the declared width W_n , and the resulting divided integer linear form

$$A_n = \frac{H_{W_n}(U_n)}{3^{R_n} 2^{S_n}}, \quad B_n = \frac{H_{W_n}(V_n)}{3^{R_n} 2^{S_n}}, \quad \rho_n = A_n F(3/2) - B_n$$

satisfies the explicit analytic condition

$$0 < |\rho_n| < \frac{1}{n}.$$

The jet equations make A_n, B_n integers. A solution would prove irrationality: if $F(3/2) = a/b$ in lowest terms, every nonzero ρ_n has absolute value at least $1/b$, contradicting the displayed bound for $n > b$. Theorem 7.7 supplies only a nonzero signed relation with the four jet equations once the pairs and size inequality are present; it does not supply primitive input rows, a nonzero combined polynomial pair, or the nonvanishing and decay of ρ_n .

Integer rescaling and a common divisor of the two specialised evaluations are the two mechanisms excluded by Corollary 7.5. Polynomial factors before specialisation, cross-row or determinant-specific arithmetic, cyclotomic factors, other additive constructions, and entirely different architectures remain outside that corollary and are not decided either way.

The remaining linear-form argument therefore has two independent gates: first find a four-jet collision that does not collapse algebraically or analytically; only then ask whether its divisibility and decay beat its height. We state those gates separately below.

12.1 Exact kernel escape

Fix, for each n , a declared width W_n and a specified family

$$(U_{n,j}, V_{n,j}, \mathcal{R}_{n,j})_{j < M_n},$$

where $U_{n,j}, V_{n,j} \in \mathbb{Z}[X]$ and $\mathcal{R}_{n,j}$ is the corresponding remainder function. Normalisation must be fixed before the jet map is formed. Call the family *polynomially primitive* when the common coefficient content of the pair in $\mathbb{Z}[X]^2$ is divided out before specialisation; as the Terminology paragraph records, that is a different operation from primitive normalisation of a specialised integer row, which is what Problem 12.1 imposes.

The specialised row content

$$g_{n,j}^{\text{spec}} = \gcd(H_{W_n}(U_{n,j}), H_{W_n}(V_{n,j}))$$

and the content of a final additive combination are separate quantities; they are not silently divided out. If either is divided out later, both the height and the remainder below are measured after that same division. Under the unit endpoint hypotheses, $g_{n,j}^{\text{spec}}$ is coprime to 6, so such division does not manufacture or destroy the 2- and 3-primary jet vanishing.

For target depths R_n, S_n , let $J_n(\lambda)$ be the four-jet signature of the pair $\sum_j \lambda_j (U_{n,j}, V_{n,j})$, and define

$$\begin{aligned} C_n &= \{\lambda \in \{-1, 0, 1\}^{M_n} \setminus \{0\} : J_n(\lambda) = 0\}, \\ K_n^{\text{poly}} &= \left\{ \lambda \in \{-1, 0, 1\}^{M_n} : \sum_j \lambda_j U_{n,j} = 0, \sum_j \lambda_j V_{n,j} = 0 \right\}, \\ K_n^{\text{rem}} &= \left\{ \lambda \in \{-1, 0, 1\}^{M_n} : \sum_j \lambda_j \mathcal{R}_{n,j}(3/2) = 0 \right\}. \end{aligned}$$

The second nullspace concerns the value at 3/2; an identically zero remainder function is a still stronger collapse and is automatically bad.

The exact pigeonhole condition is

$$2^{M_n} > 3^{2R_n} 2^{2S_n}, \quad \text{equivalently} \quad M_n > 2R_n \log_2 3 + 2S_n. \quad (8.1)$$

Thus the least admissible integer rank is

$$M_{\min}(R, S) = \lfloor 2R \log_2 3 + 2S \rfloor + 1.$$

The checked condition $M \geq 4R + 2S$ for $R > 0$ is a convenient sufficient corollary, not the exact threshold. Moreover, with $N = 2^{M_n}$ and $Q = 3^{2R_n} 2^{2S_n}$, convexity of the fibre sizes gives at least

$$\frac{1}{2} \left(\frac{N^2}{Q} - N \right) \quad (8.2)$$

unordered equal-signature selector pairs. A normality estimate can therefore win by showing that fewer than this many collision pairs land in the two bad nullspaces.

Problem 12.2 (four-jet kernel escape). For one literal polynomially primitive family satisfying (8.1), prove, preferably by comparing the lower bound (8.2) with the bad-pair multiplicities, that

$$C_n \setminus (K_n^{\text{poly}} \cup K_n^{\text{rem}}) \neq \emptyset$$

for all sufficiently large n .

Theorem 7.7 supplies only $C_n \neq \emptyset$. It gives a nonzero selector difference, but it gives neither polynomial-pair nonvanishing nor remainder nonvanishing. Problem 12.2 is finite algebra and normality; it makes no asymptotic product-formula claim. It is the first of the two gates in Problem 12.1: it asks for the two nonvanishing conclusions, and not for the analytic condition stated there.

12.2 Determinant families

An exploratory calculation suggests a contiguous a_0 -shift determinant at $r_n = 13n + 2$, after apparent failure of lower ranks $r \leq 13n + 1$. These computations are not Lean theorems. More importantly, no literal definition of the matrix $A_{n,r}$ is given here, so writing merely “ $\det A_{n,13n+2} \neq 0$ ” would not yet be a public mathematical question.

Problem 12.3 (the first saturated determinant). Give the entrywise formula for the contiguous a_0 -shift matrix $A_{n,r}$, the exact relation $M_n = M(n, r)$ between its size and the number of coefficient pairs, and the declared width W_n . Then determine whether

$$\det A_{n,13n+2} \neq 0$$

for every sufficiently large n . At the first computable scales, report the rank, determinant, coefficient-pair contents, specialised row contents, four endpoint jets, and the first nonzero remainder coefficient. A negative answer must identify a systematic rank relation or the first exact failing n .

This formulation makes the matrix definition part of the problem statement rather than relying on undeclared notation.

Problem 12.4 (minimal non- a_0 deformation). If the saturated contiguous family collapses, enlarge it by exactly one a_1 -shift while retaining the same source cone and widths. Does this one-direction extension increase the polynomial-pair rank and produce a collision outside $K_n^{\text{poly}} \cup K_n^{\text{rem}}$? If every such one-shift extension collapses, prove that class-wide obstruction before adding a second new direction.

This is the smallest specified deformation beyond the contiguous family; it replaces the unbounded request for a “genuinely independent deformation.”

12.3 Asymptotic adequacy after kernel escape

Suppose a good collision has been found. Let $D_n = 3^{R_n} 2^{S_n}$ be the certified local divisor (or replace it by the exact determinant-specific divisor), let H_n be the coefficient or exterior height after precisely the normalisation just declared, and let $L_n \neq 0$ be the resulting analytic form or exterior remainder.

Problem 12.5 (negative normalised product-formula margin). Prove the explicit estimate

$$\limsup_{n \rightarrow \infty} \frac{\log H_n + \log |L_n| - R_n \log 3 - S_n \log 2}{n^2} < 0. \quad (8.3)$$

Every denominator, row content and final-combination content must already be included in H_n and L_n .

Nonvanishing alone does not address (8.3). Conversely, excellent formal decay is irrelevant if every jet collision lies in a bad nullspace. The exploratory adjacent-exterior calculation leaves a positive normalised exponent of about $110.850 n^2$; this is not a checked theorem. Any proposed improvement must remove this explicit deficit; extra divisibility alone is insufficient.

12.4 One exact alternative-criterion test

A separate possible method is Mahler's method. Its first applicability test has an exact negative answer, which we record here; an earlier version of this note left it open.

Proposition 12.6 (no finite simultaneous 2/3-system). *Let*

$$\mathcal{L}(z) = \sum_{n \geq 1} \frac{z^n}{1 - z^n}.$$

There is no finite-dimensional $\mathbb{Q}(z)$ -vector space that contains $\mathcal{L}$ and is stable under both $z \mapsto z^2$ and $z \mapsto z^3$.

Proof. Suppose V were such a space, of dimension d . Stability under $z \mapsto z^2$ places the $d+1$ elements $\mathcal{L}(z), \mathcal{L}(z^2), \dots, \mathcal{L}(z^{2^d})$ in V , so they are linearly dependent over $\mathbb{Q}(z)$; clearing denominators gives polynomials $P_0, \dots, P_d$, not all zero, with $\sum_{i=0}^d P_i(z) \mathcal{L}(z^{2^i}) = 0$. Thus $\mathcal{L}$ is 2-Mahler, and the same argument under $z \mapsto z^3$ makes it 3-Mahler. Since 2 and 3 are multiplicatively independent, a theorem of Adamczewski and Bell [12, Theorem 1] then forces $\mathcal{L}$ to be a rational function. That contradicts Rivin's periodic-coefficient corollary [17, Cor. 6.4, p. 9], by which $\mathcal{L}$ is not rational. $\square$

The proposition uses no property of the point 2/3: the obstruction is functional and appears before regularity at a particular point is considered. It closes the simultaneous route only. It says nothing about a single k -Mahler system, about q -difference or Mahler-type arguments that do not require stability under two multiplicatively independent substitutions, or about special-value theorems reached by other means. Neither ingredient is proved here; both are cited.

12.5 The quantitative height limitation

The module `HermitePadeNoGo` defines an explicit rectangular two-parameter exponent model. Writing $\sigma = 1 + \rho + u$, its denominator-cleared gap has the exact expansion

$$-\pi^2 \rho^2 - \pi^2 \rho u - 2\pi^2 \rho - 2\rho^2 - 10\rho u - 4\rho - 6u^2 - 8u.$$

Thus for $\rho \geq 0$ and $\sigma \geq 1 + \rho$ the cleared gap is nonpositive, and it vanishes exactly when $\rho = 0$ and $\sigma = 1$ ([exact expansion](#), [nonpositivity](#), [equality case](#)). Equivalently, within that model the displayed threshold never exceeds the classical one-function margin, with equality only at the classical endpoint ([bound](#), [equality](#)). This is an algebraic theorem about the four defined exponent expressions and only that explicit rectangular model. It does not construct polynomials, remainders, integrality, a determinant, or asymptotics, and it is not a method-universal no-go theorem. The separate 81/200 cutoff of the preceding section satisfies

$$\frac{81}{200} < \frac{\log 2}{\log 3}.$$

Thus “improve the height theorem” has a precise numerical target.

Problem 12.7 (optimal admissible height threshold). Specify a concrete class A of primitive, noncollapsed constructions, excluding scalar row rescaling, proportional permutation-orbit forms, pure row-content gain, and constructions governed by the rectangular exponent model above. For that class define

$$\theta_* = \sup_{\alpha \in A} \frac{C_0(\alpha)}{C_1(\alpha)}.$$

Prove one of the following: $\theta_* > 81/200$ by an explicit construction; an exact value for θ_* ; a converse $\theta_* < \log 2 / \log 3$; or a class theorem showing that every member reduces to one of the checked scalar or rectangular no-go mechanisms.

Reaching $3/2$ requires a threshold strictly beyond $\log 2 / \log 3 \approx 0.6309$; exceeding $81/200 = 0.405$ is insufficient. The unrestricted question of which rational bases give irrational values remains open and is not reduced to any one of these problems.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision `ca0e13f8acf5`. The declarations of this note live in seven modules: `RationalBaseLambert`, `QAperyDiagonalNonEquivalence`, `RationalPadeArithmetic`, `ZudilinConeArithmetic`, `ZudilinHeightRegion`, `HermitePadeNoGo`, and `BezoutPluckerJets`. The first contains the corridor, cleared-tail recurrence, and elementary $7/2$ certificate; the second checks the finite

$n = 0$ diagonal residual; the remaining four separate the Padé exponent arithmetic, end-point arithmetic, logarithmic comparisons, rectangular exponent model, and Bézout–Plücker tail collapse. The link coordinates are validated against that pinned revision, so they remain correct as later work moves lines in the working tree.

References

- [1] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. (N.S.) **12** (1948), 63–66.
- [2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.
- [3] P. B. Borwein, *On the irrationality of $\sum 1/(q^n + r)$* , J. Number Theory **37** (1991), no. 3, 253–259, doi:10.1016/S0022-314X(05)80041-1.
- [4] P. B. Borwein, *On the irrationality of certain series*, Math. Proc. Cambridge Philos. Soc. **112** (1992), no. 1, 141–146, doi:10.1017/S030500410007081X. Van Assche cites Lemma 2 for the neighbouring little- q -Legendre evaluation used above.
- [5] T. Amdeberhan and D. Zeilberger, *q -Apéry irrationality proofs by q -WZ pairs*, Adv. Appl. Math. **20** (1998), no. 2, 275–283, arXiv:math/9804122, doi:10.1006/aama.1997.0565.
- [6] P. Bundschuh and K. Väänänen, *Arithmetical investigations of a certain infinite product*, Compositio Math. **91** (1994), no. 2, 175–199.
- [7] D. Duverney, *À propos de la série $\sum_{n=1}^{+\infty} x^n/(q^n - 1)$* , J. Théor. Nombres Bordeaux **8** (1996), no. 1, 173–181. Théorème 2, printed p. 174, gives irrationality of $\sum_{n \geq 1} (q^n - 1)^{-1}$ for $q = r/s$ under $\log |s|/\log |r| < \frac{1}{3}(1 - 3/\pi^2)$.
- [8] W. Zudilin, *Heine’s basic transform and a permutation group for q -harmonic series*, Acta Arith. **111** (2004), no. 2, 153–164, doi:10.4064/aa111-2-4.
- [9] W. Zudilin, *Irrationality of the generalized q -logarithm*, Res. Number Theory **2** (2016), Paper 24, doi:10.1007/s40993-016-0042-x. The remark that the results extend to non-integer $p = r/s$, $|p| > 1$, under an assumption $\log |r| > c \log |s|$ for a computable $c > 0$, is at the end of the introduction; no value of c is computed there.
- [10] T. Matala-aho, K. Väänänen and W. Zudilin, *New irrationality measures for q -logarithms*, Math. Comp. **75** (2006), no. 254, 879–889. Integer base throughout; p. 880 records that the methods there do not sharpen the exponent of [8].
- [11] S. Chern, L. Jiu, S. Li, and L. Wang, *Leading coefficient in the Hankel determinants related to binomial and q -binomial transforms*, Adv. Appl. Math. **176** (2026), 103051, doi:10.1016/j.aam.2026.103051.
- [12] B. Adamczewski and J. P. Bell, *A problem about Mahler functions*, Ann. Sc. Norm. Super. Pisa Cl. Sci. **17** (2017), no. 4, 1301–1355; arXiv:1303.2019, 2013. Theorem 1: over a field of characteristic zero, a power series is both k - and ℓ -Mahler for multiplicatively independent k, ℓ if and only if it is a rational function.
- [13] G. Rhin and C. Viola, *On a permutation group related to $\zeta(2)$* , Acta Arith. **77** (1996), no. 1, 23–56, doi:10.4064/aa-77-1-23-56.
- [14] W. Van Assche, *Little q -Legendre polynomials and irrationality of certain Lambert series*, Ramanujan J. **5** (2001), no. 3, 295–310, doi:10.1023/A:1012930828917.

- [15] J. Vandehey, *On an incomplete argument of Erdős on the irrationality of Lambert series*, *Integers* **13** (2013), Paper A58.
- [16] F. Luca and Y. Tachiya, *Linear independence results for the values of divisor functions series*, *RIMS Kôkyûroku No. 2014* (2017), 138–150. Theorem A on p. 139 restates the periodic-coefficient irrationality theorem; Example 1 on p. 140 gives the divisor-function specialization.
- [17] I. Rivin, *Zero Coefficients of Rational Power Series and Rational Lambert Series*, arXiv:2604.25151v1, 2026. Theorem 1.1 is on p. 2 and proved on pp. 6–7; the periodic-coefficient Corollary 6.4 is on p. 9.
- [18] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, *Acta Math. Hungar.* **175** (2025), 572–608; arXiv:2406.17593, 2024.
- [19] T. F. Bloom, *Erdős Problem #1049*, erdosproblems.com/1049, accessed 28 July 2026 (page displays “last edited 28 September 2025”). The current record labels the problem open, cites [Er88c, p. 102] and [Er48], and explicitly describes its status as the website owner’s present assessment rather than a literature-completeness guarantee.
- [20] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.1049*, Lean source at commit f776d2f, 2026, accessed 28 July 2026. The irrationality declarations are unproved; the Lambert-series identity is proved.